Group 1

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Class Diagram for StoreSystem, Client, Product, and WishlistItem

(1) a table like the one shown in Chapter 5, (Table 5.1) explaining how the methods were found in the sequence diagrams.

The methods in these tables were identified from the messages and responsibilities shown in the sequence diagrams. Each message sent to an object represents an operation that the receiving software class must be able to perform. The methods were therefore assigned to StoreSystem, Client, Product, and WishlistItem according to the responsibilities established in the design. The query methods were also included because the system must display all clients, all products, and the products and quantities contained in a specified client's wishlist.

StoreSystem

Sequence Diagram Extraction	Method Identification	Task Performance
Add Client	addClient(name: String, address: String): Client	Creates a new Client and adds it to the collection of clients.
Add Client	generateClientID(): int	Generates a unique ID for the new Client.
Add Products	addProduct(name: String, quantityInStock: int, salePrice: double): Product	Creates a new Product and adds it to the collection of products.
Add Products	generateProductID(): int	Generates a unique ID for each new Product.
Add Products to a Client Wishlist	addToWishlist(clientID: int, productID: int, quantity: int): boolean	Coordinates adding a Product to a Client's wishlist or changing the quantity of an existing WishlistItem.
Add Products to a Client Wishlist	findClient(clientID: int): Client	Searches for and returns the Client identified by the client ID.
Add Products to a Client Wishlist	findProduct(productID: int): Product	Searches for and returns the Product identified by the product ID.
Display All Clients With Details	getAllClients(): List<Client>	Returns all Clients so their details can be displayed.
Display All Products With Details	getAllProducts(): List<Product>	Returns all Products so their details can be displayed.
Display Client Wishlist	getWishlistItems(clientID: int): List<WishlistItem>	Returns the specified Client's WishlistItems so the Products and desired quantities can be displayed.

Client

Sequence Diagram Extraction	Method Identification	Task Performance
Add Client	Client(clientID: int, name: String, address: String)	Creates a Client and stores the Client's ID, name, and address.
Add Products to a Client Wishlist	findWishlistItem(productID: int): WishlistItem	Searches the Client's wishlist for the WishlistItem associated with the specified Product.
Add Products to a Client Wishlist	addWishlistItem(product: Product, quantity: int): boolean	Creates and adds a WishlistItem containing the specified Product and desired quantity.
Display Client Wishlist	getWishlistItems(): List<WishlistItem>	Returns all WishlistItem objects belonging to the Client.
Display All Clients With Details	getDetails(): String	Returns the Client's stored information so it can be displayed.

Product

Sequence Diagram Extraction	Method Identification	Task Performance
Add Products	Product(productID: int, name: String, quantityInStock: int, salePrice: double)	Creates a Product and stores its ID, name, quantity in stock, and sale price.
Display All Products With Details	getDetails(): String	Returns the Product's stored information so it can be displayed.
Display Client Wishlist	getDetails(): String	Returns the Product information needed when displaying the contents of a Client's wishlist.

WishlistItem

Sequence Diagram Extraction	Method Identification	Task Performance
Add Products to a Client Wishlist	WishlistItem(product: Product, quantity: int)	Creates a new WishlistItem containing a reference to the Product and the desired quantity.
Add Products to a Client Wishlist	setQuantity(quantity: int): void	Replaces the existing quantity when the Product is already in the Client's wishlist.
Display Client Wishlist	getProduct(): Product	Returns the Product referenced by the WishlistItem.

Display Client Wishlist	getQuantity(): int	Returns the desired quantity stored in the WishlistItem.

(2) A class diagram like the ones shown in Fig 5.13

